

1 Knowledge Base Deteksi Anomali Tanda Vital (Dewasa) – RAG Ready

1.1 1. Body Temperature

Parameter: temp

Unit: °C

Normal Range:

- 36.5 – 37.5

Abnormal Conditions:

- Hypothermia: < 36.5
- Fever: > 38
- Hyperpyrexia: > 40

Severity Classification:

- Mild: 37.6 – 38
- Moderate: 38 – 40
- Critical: > 40

1.2 2. Heart Rate

Parameter: hr

Unit: bpm

Normal Range:

- 60 – 100

Abnormal Conditions:

- Bradycardia: < 60
- Tachycardia: > 100
- Severe Bradycardia: < 40
- Severe Tachycardia: > 140

Severity Classification:

- Mild: 100 – 120
- Moderate: 120 – 140
- Critical: > 140

1.3 3. Respiratory Rate

Parameter: rr

Unit: breaths/min

Normal Range:

- 12 – 20

Abnormal Conditions:

- Bradypnea: < 12
- Tachypnea: > 20
- Apnea: 0

Severity Classification:

- Mild: 20 – 24
- Moderate: 25 – 30
- Critical: > 30

1.4 4. Blood Pressure – Systolic

Parameter: sbp

Unit: mmHg

Normal Range:

- 90 – 120

Abnormal Conditions:

- Hypotension: < 90
- Elevated: 120 – 129
- Hypertension Stage 1: 130 – 139
- Hypertension Stage 2: ≥ 140
- Hypertensive Crisis: ≥ 180

Severity Classification:

- Mild: 120 – 129
- Moderate: 130 – 139
- Severe: 140 – 179
- Critical: ≥ 180

1.5 5. Blood Pressure – Diastolic

Parameter: dbp

Unit: mmHg

Normal Range:

- 60 – 80

Abnormal Conditions:

- Hypotension: < 60
- Hypertension Stage 1: 80 – 89
- Hypertension Stage 2: ≥ 90
- Hypertensive Crisis: ≥ 120

Severity Classification:

- Mild: 80 – 89
- Moderate: 90 – 99
- Severe: 100 – 119
- Critical: ≥ 120

1.6 6. Oxygen Saturation

Parameter: spo2

Unit: %

Normal Range:

- 95 – 100

Abnormal Conditions:

- Mild Hypoxia: 90 – 94
- Moderate Hypoxia: 85 – 89
- Severe Hypoxia: < 85

Severity Classification:

- Mild: 90 – 94
- Moderate: 85 – 89
- Critical: < 85

2 Clinical Rules

2.1 1. Hypotension Detection

- $sbp < 90$
- OR $dbp < 60$

2.2 2. Hypertension Stage 1

- $sbp\ 130 - 139$
- OR $dbp\ 80 - 89$

2.3 3. Hypertension Stage 2

- $sbp \geq 140$
- OR $dbp \geq 90$

2.4 4. Hypertensive Crisis

- $sbp \geq 180$
- OR $dbp \geq 120$

2.5 5. Shock Detection

- $sbp < 90$
- AND $hr > 100$

2.6 6. Sepsis Detection

- $temp > 38$
- $hr > 100$
- $sbp < 90$

2.7 7. Respiratory Failure

- $rr > 24$
- $spo2 < 90$